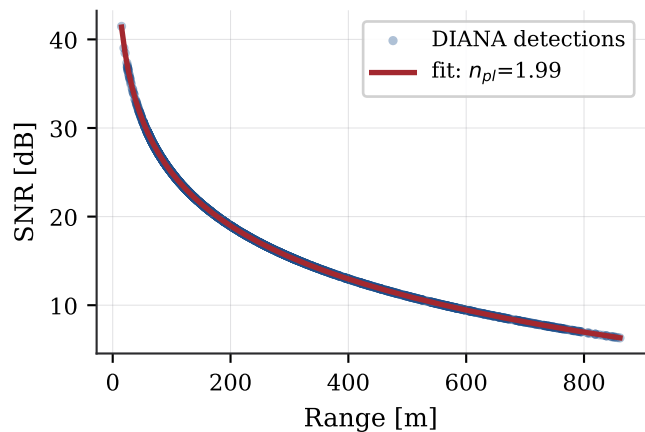
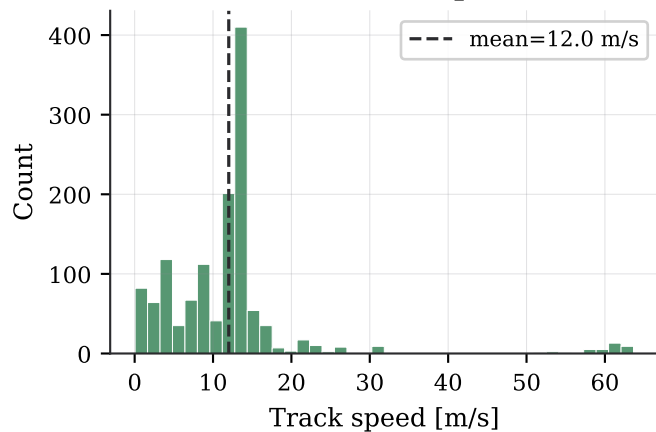


Figure 1 — Real-World Sensor & Telemetry Data Underlying the Simulation Calibration

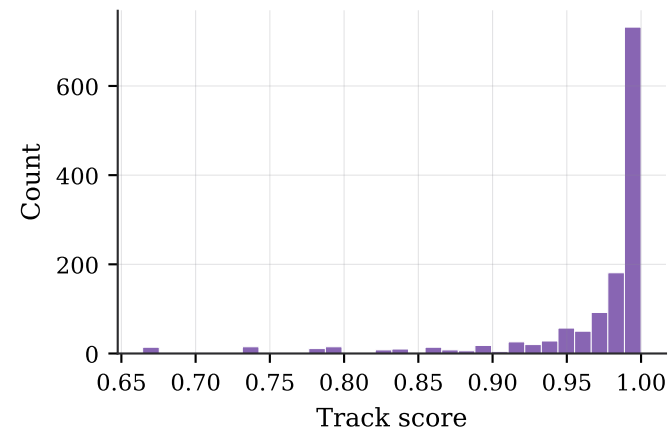
(a) Radar Path-Loss Fit (DIANA RF, real data)



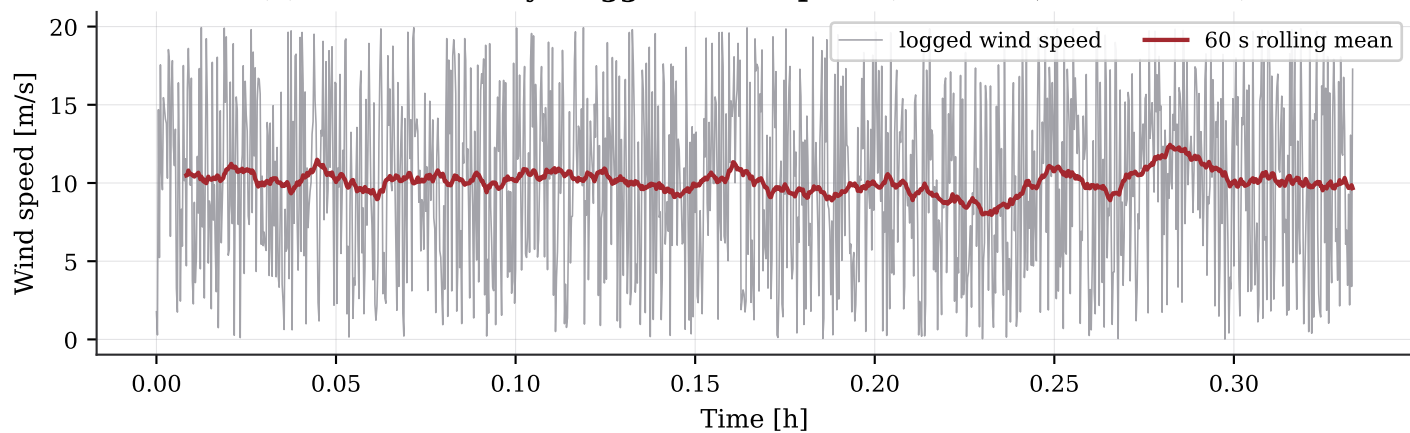
(b) ALVIRA Radar Track Speeds (real data)



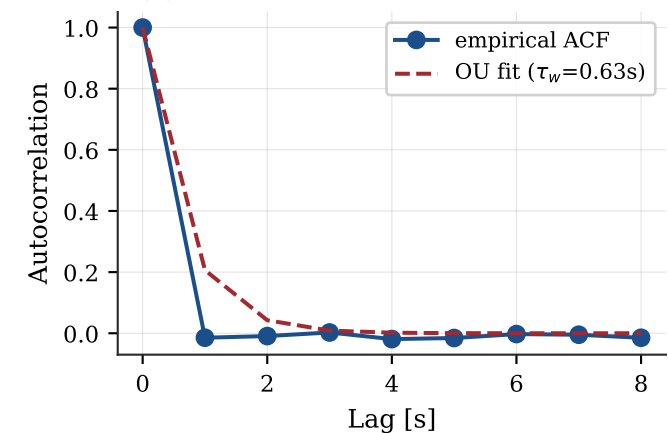
(c) ALVIRA Track Confidence Score (real data)



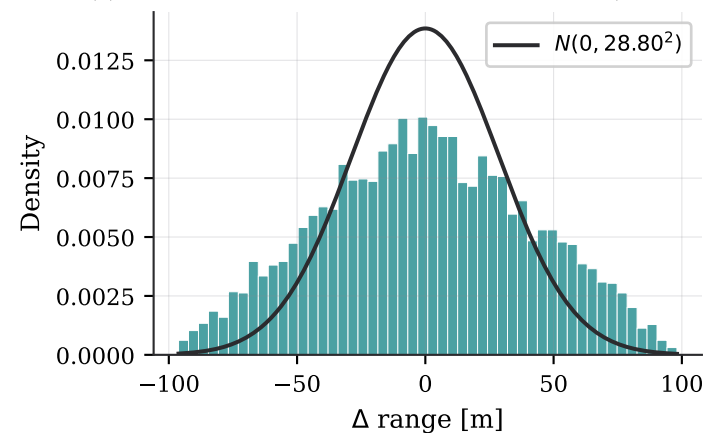
(d) UAV Telemetry: Logged Wind Speed (real data, first 20 min)



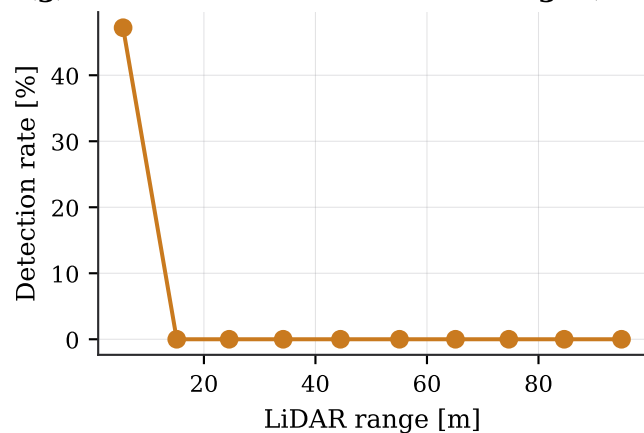
(e) Wind OU-Process Calibration



(f) LiDAR Increment Noise Floor (real data)



(g) Detection Rate vs LiDAR Range (deciles)



(h) DIANA Classif. Confidence by Target Type

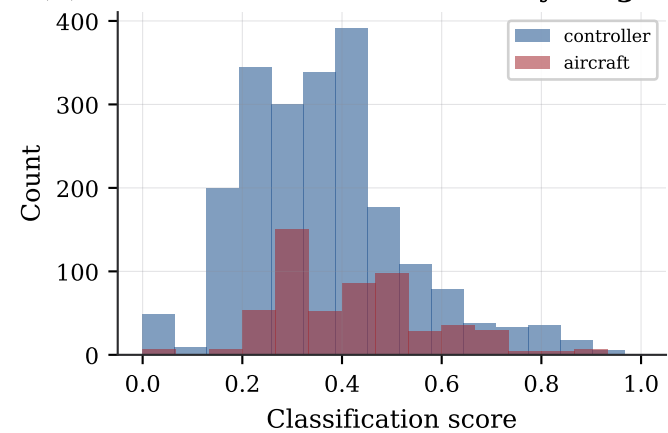
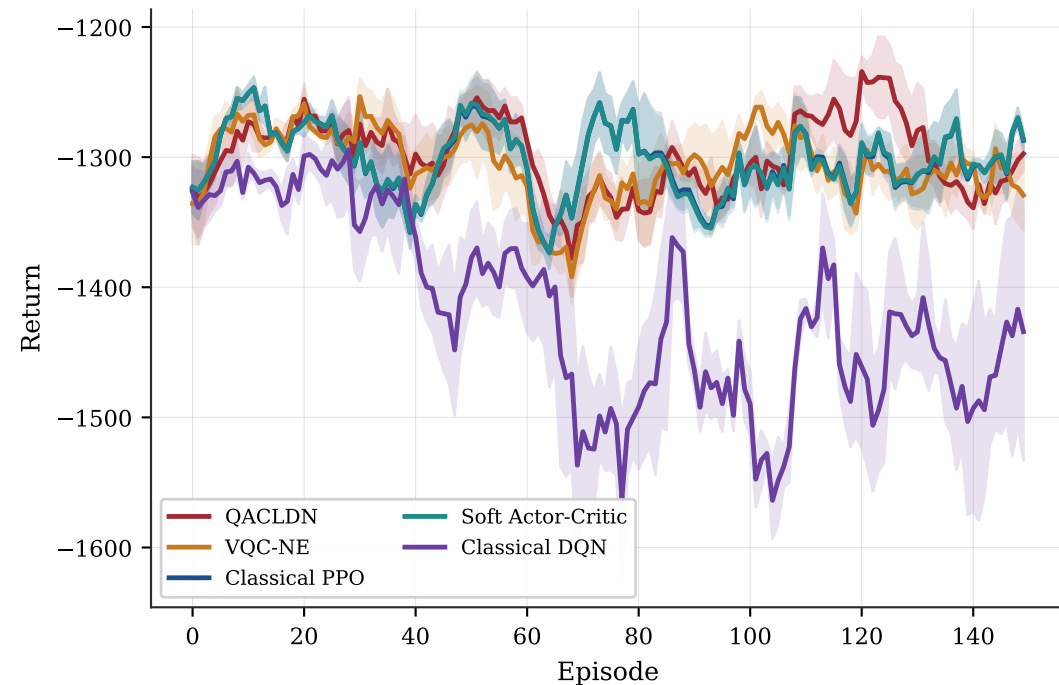
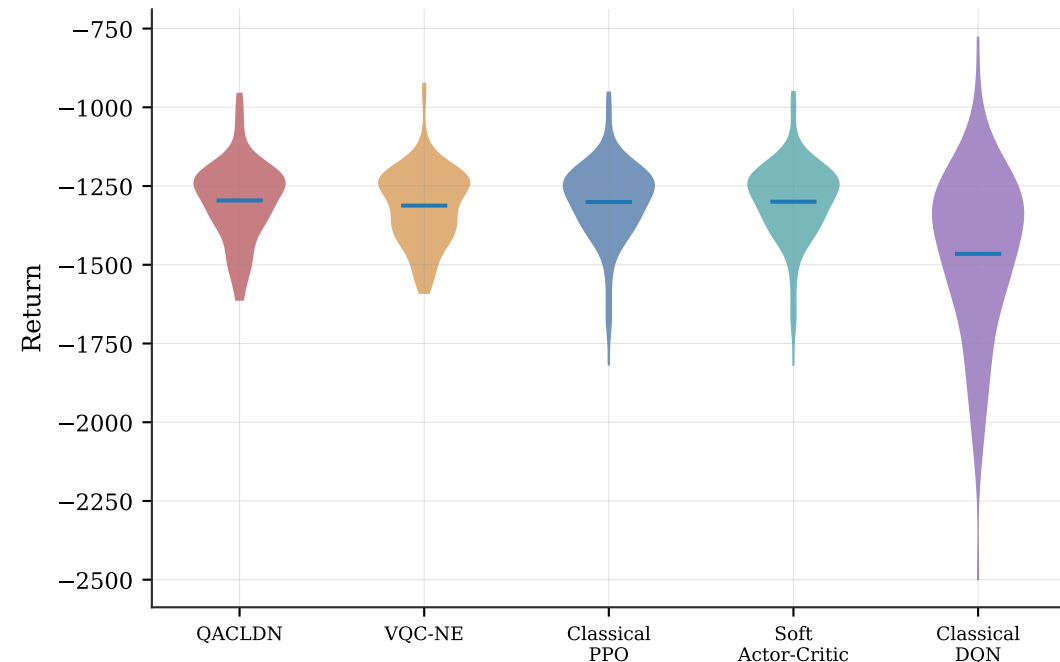


Figure 2 — Learning Dynamics: QACLDN vs. Classical & Ablation Baselines
(3 seeds x 150 episodes; shaded band = ± 1 s.e.m. across seeds)

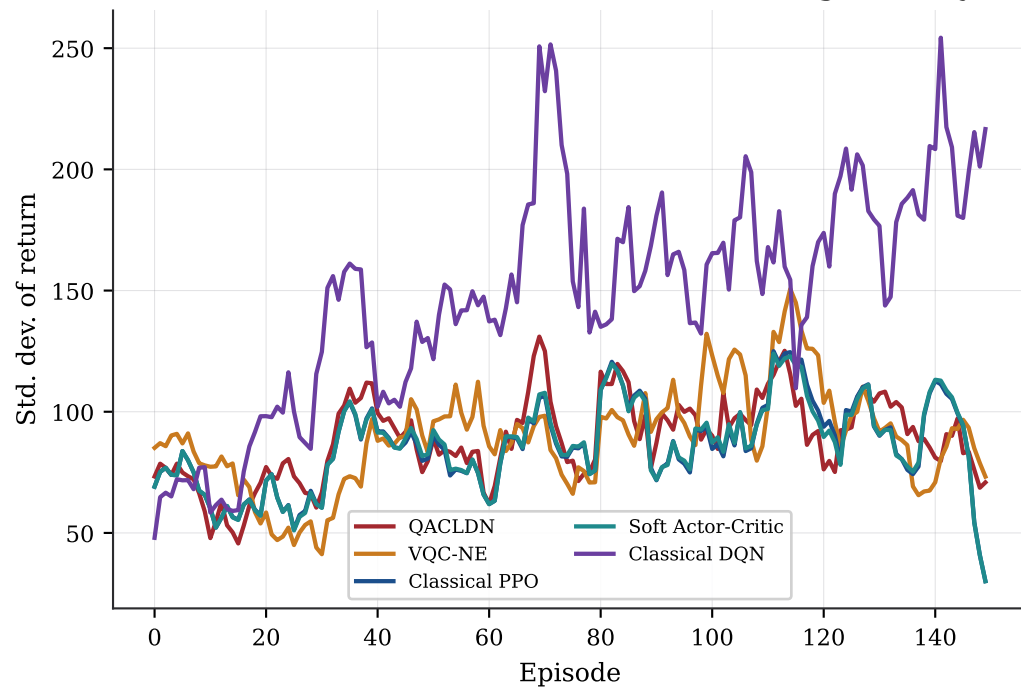
(a) Smoothed Episodic Return



(b) Return Distribution — Final 20% of Training



(c) Cross-Seed Std. Dev. of Return (training stability)



(d) Improvement Over Own Initial Performance

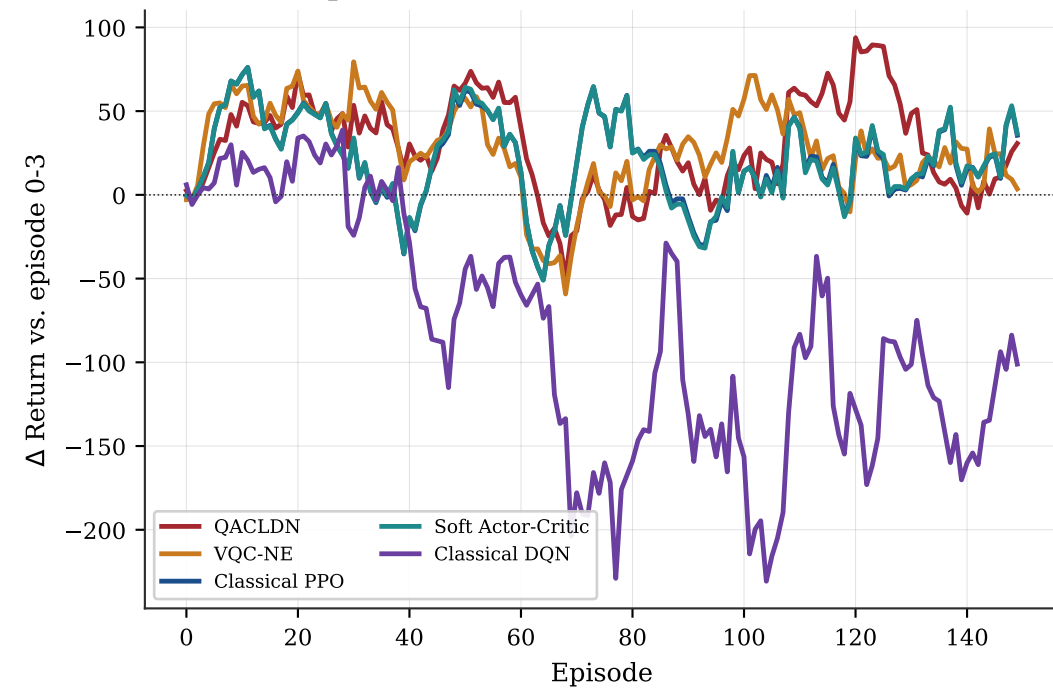
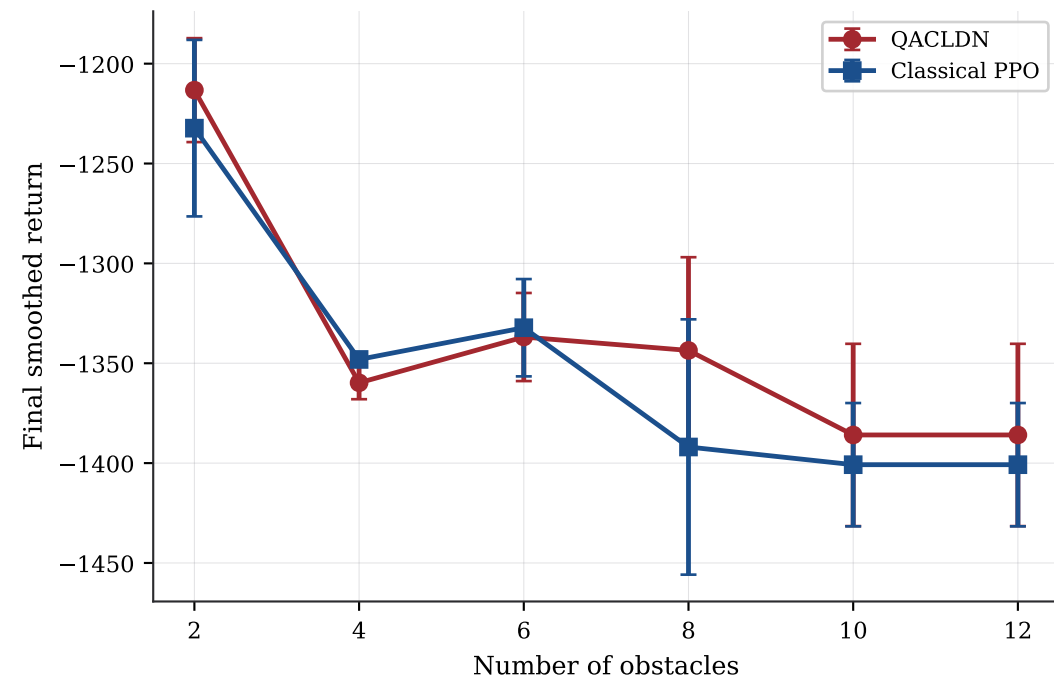
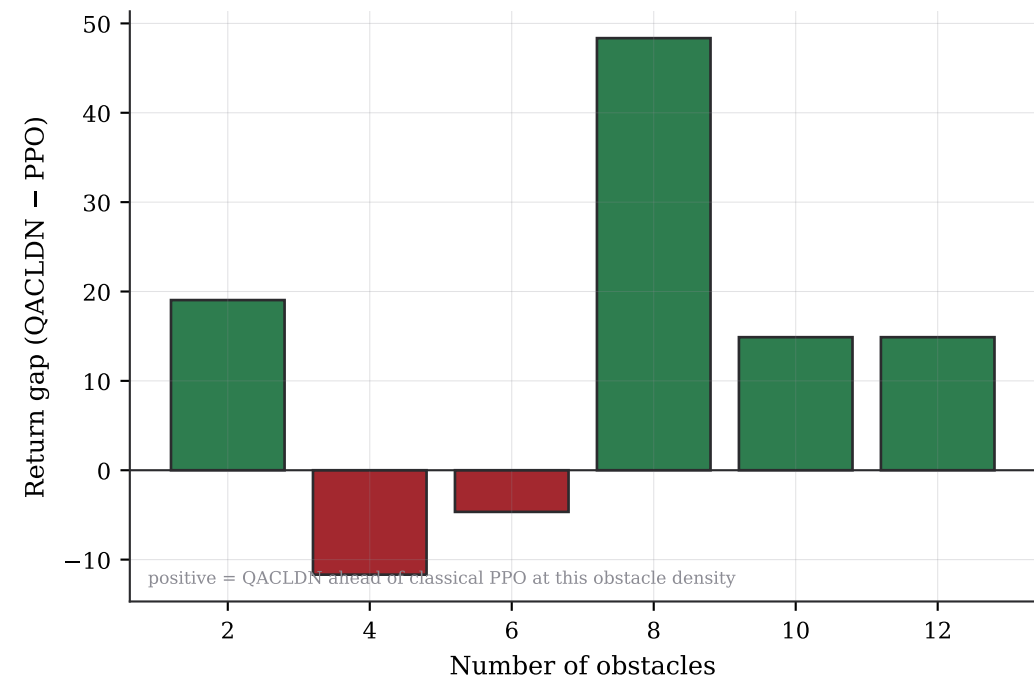


Figure 3 — Obstacle-Density Scalability and Qubit-Count Sample Complexity

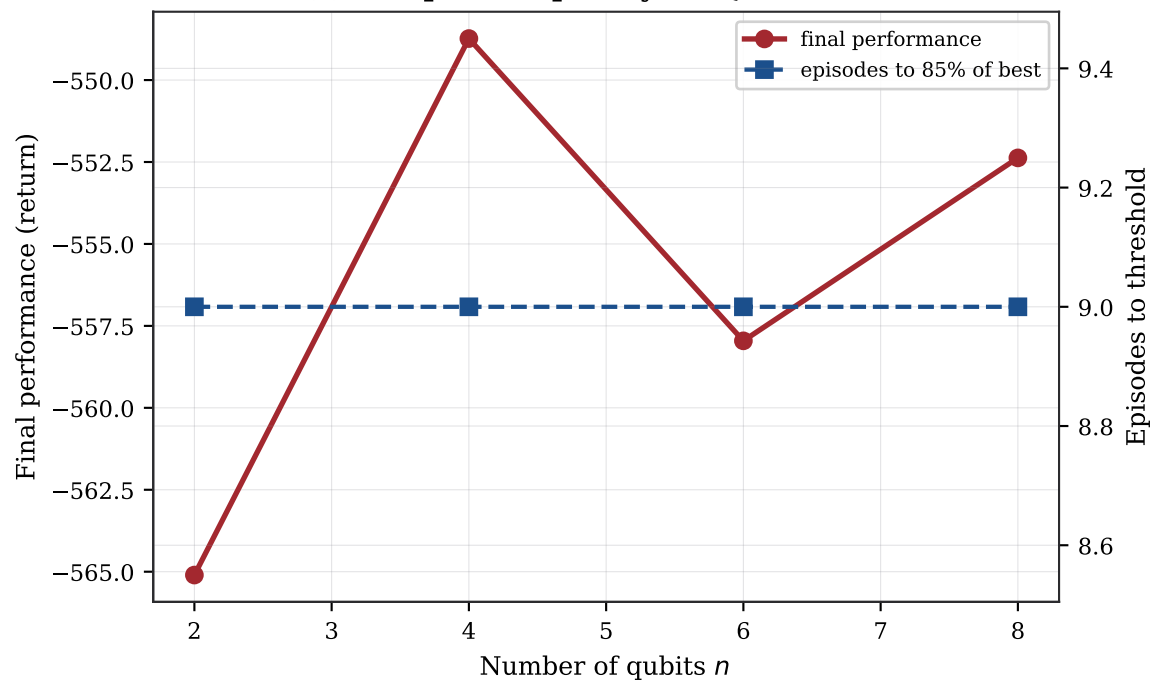
(a) Final Performance vs. Obstacle Count



(b) QACLDN – Classical PPO Final Return, by Obstacle Count



(c) Sample Complexity vs. Qubit Count



(d) Total Trainable Parameters vs. Qubit Count

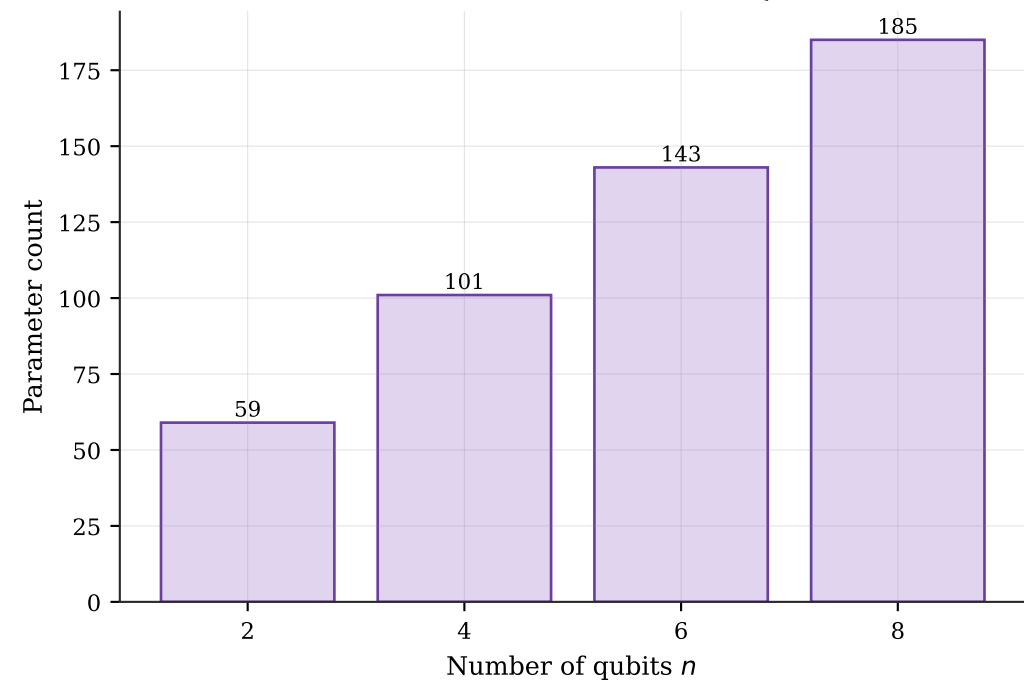


Figure 4 — Circuit-Design Performance Map & Robustness Under NISQ Noise / Wind

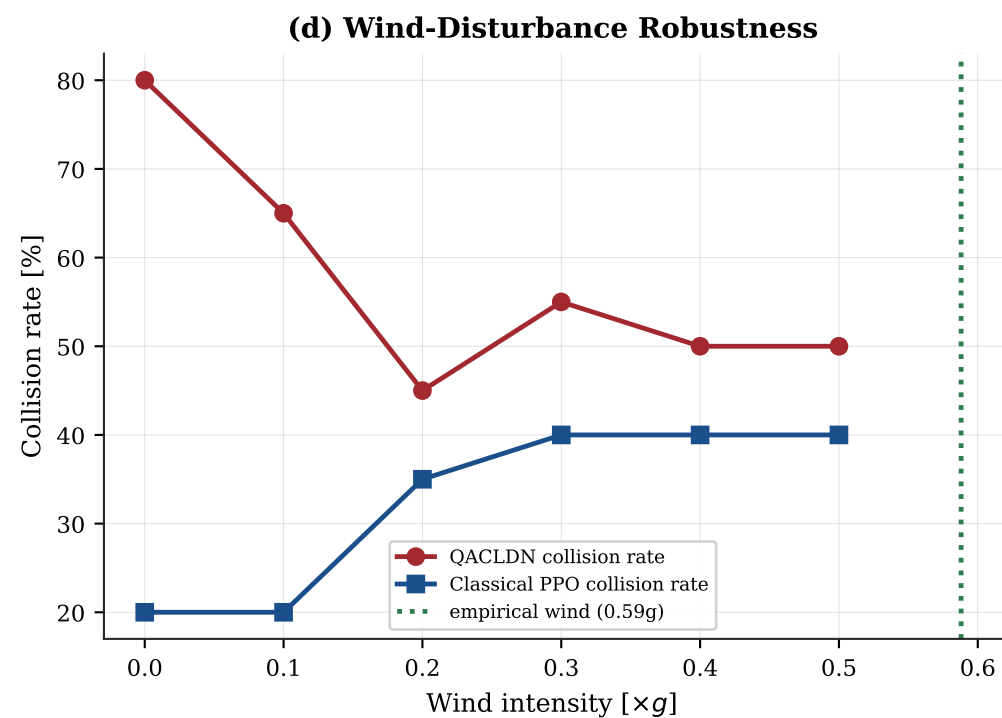
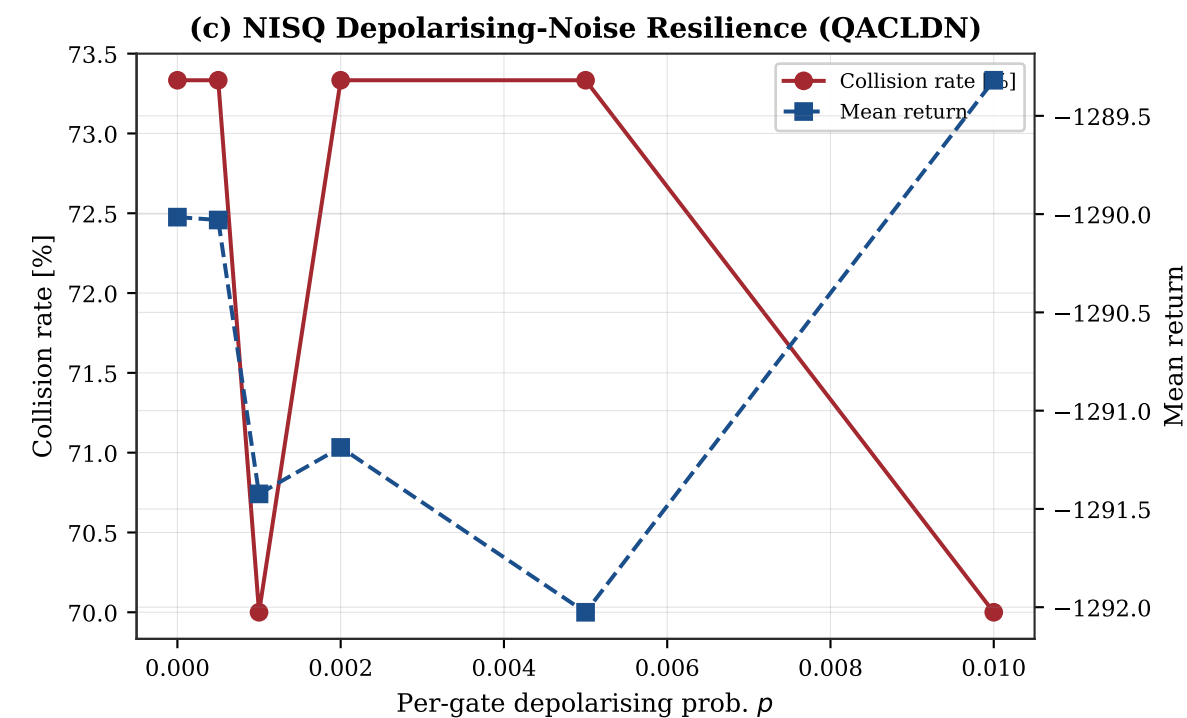
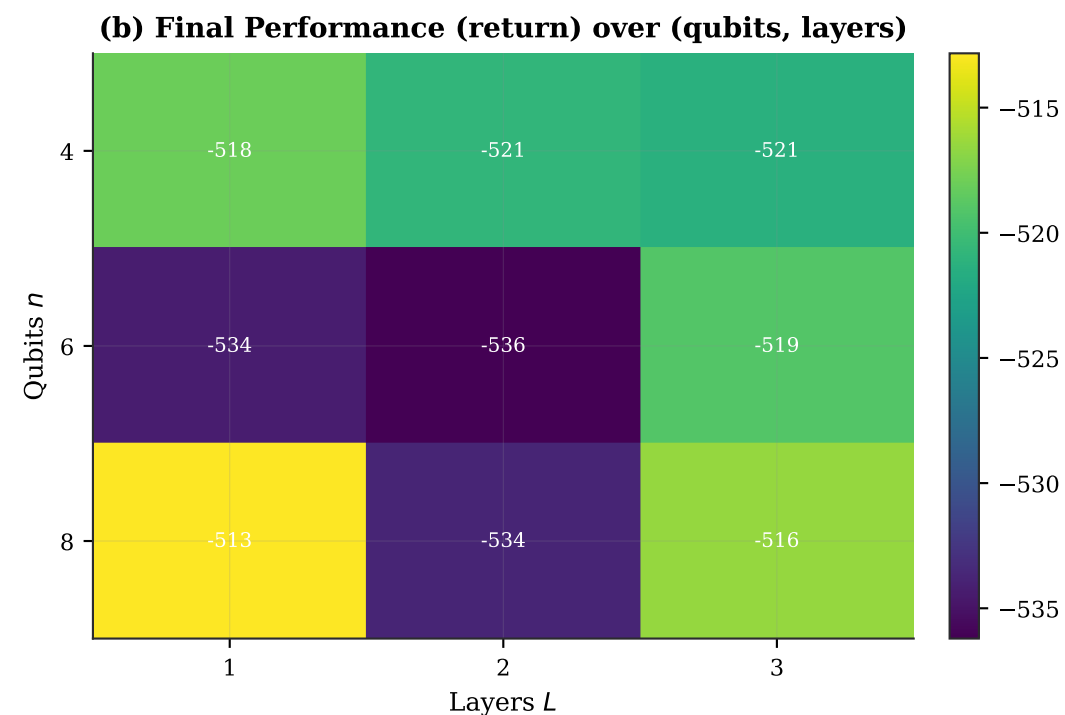
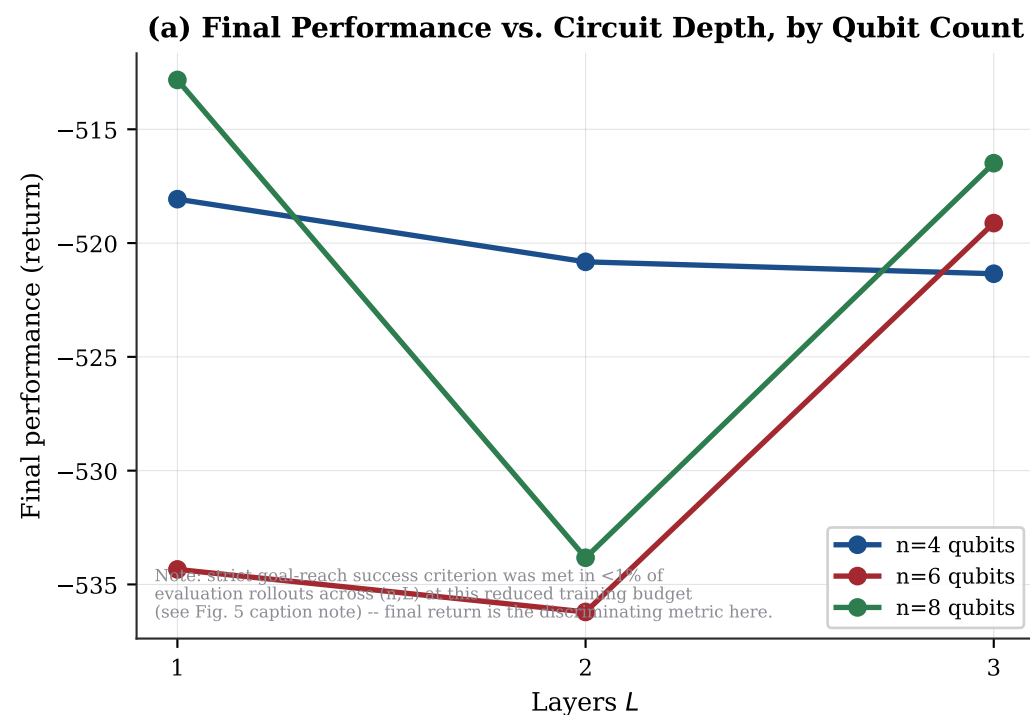
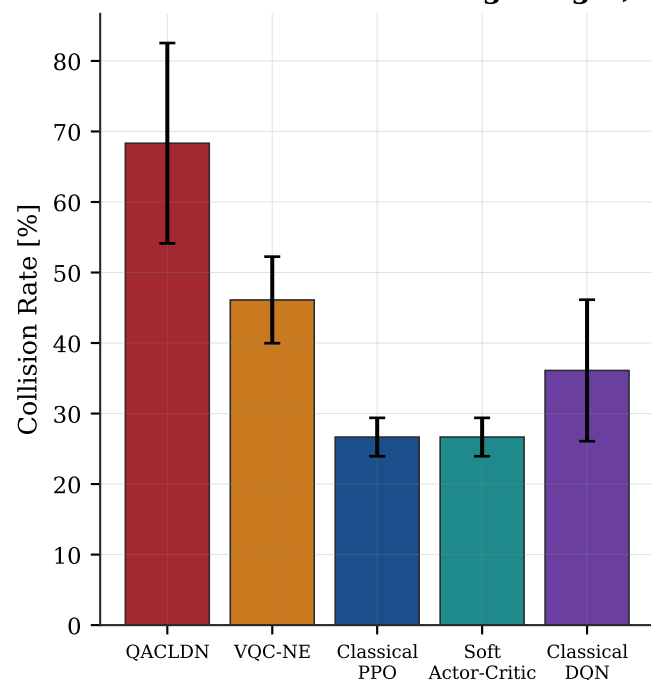
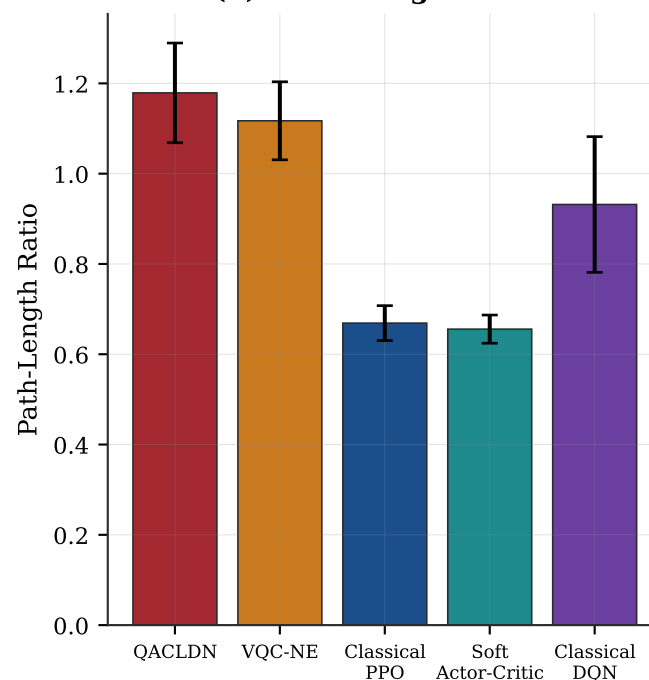


Figure 5 — Final Trajectory-Quality Metrics & Representative Flight Paths

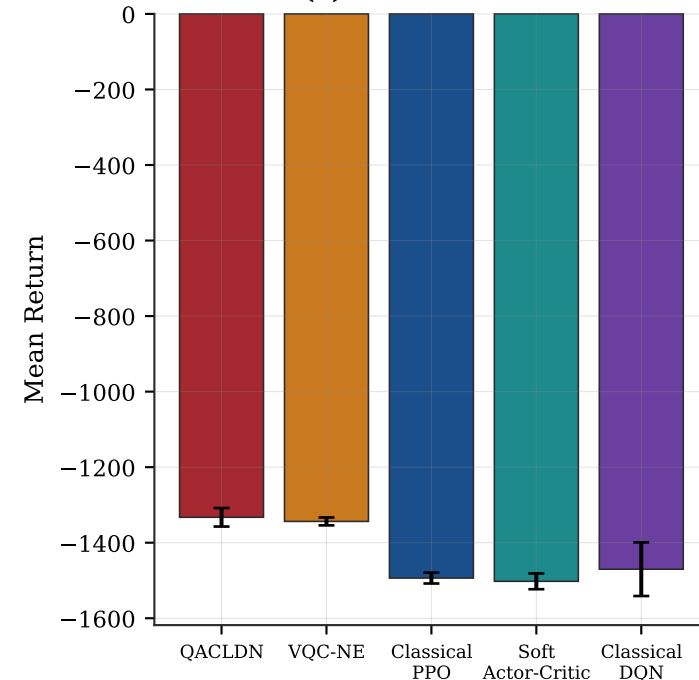
(a) Collision Rate [%]
(strict success rate was ~0% for all methods at this training budget)



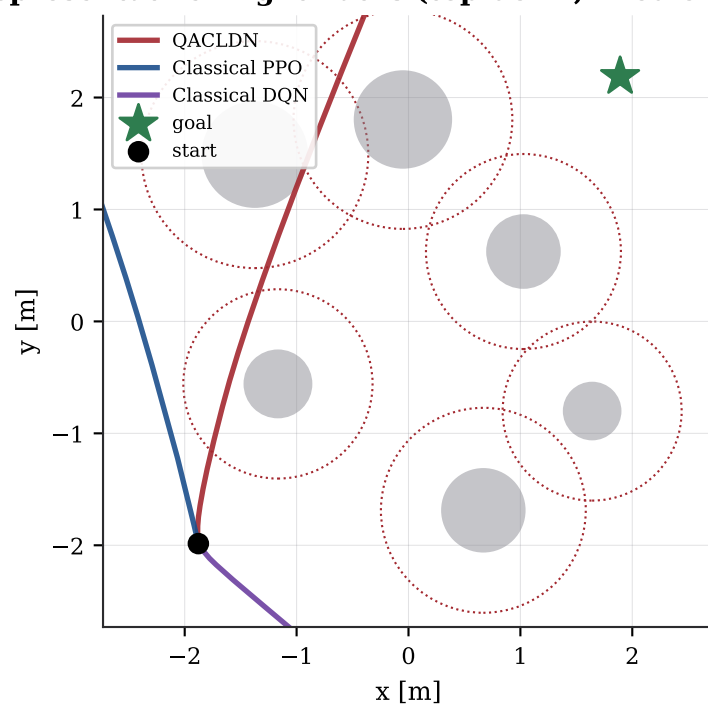
(b) Path-Length Ratio



(c) Mean Return



(d) Representative Flight Paths (top-down, fixed obstacle field)



(e) Energy-Efficiency vs. Mean Return

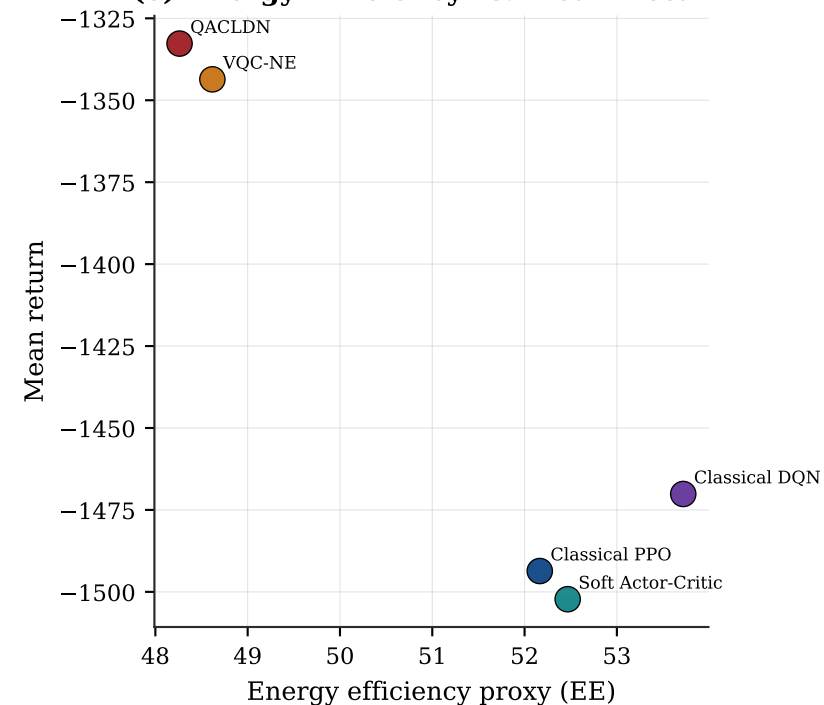
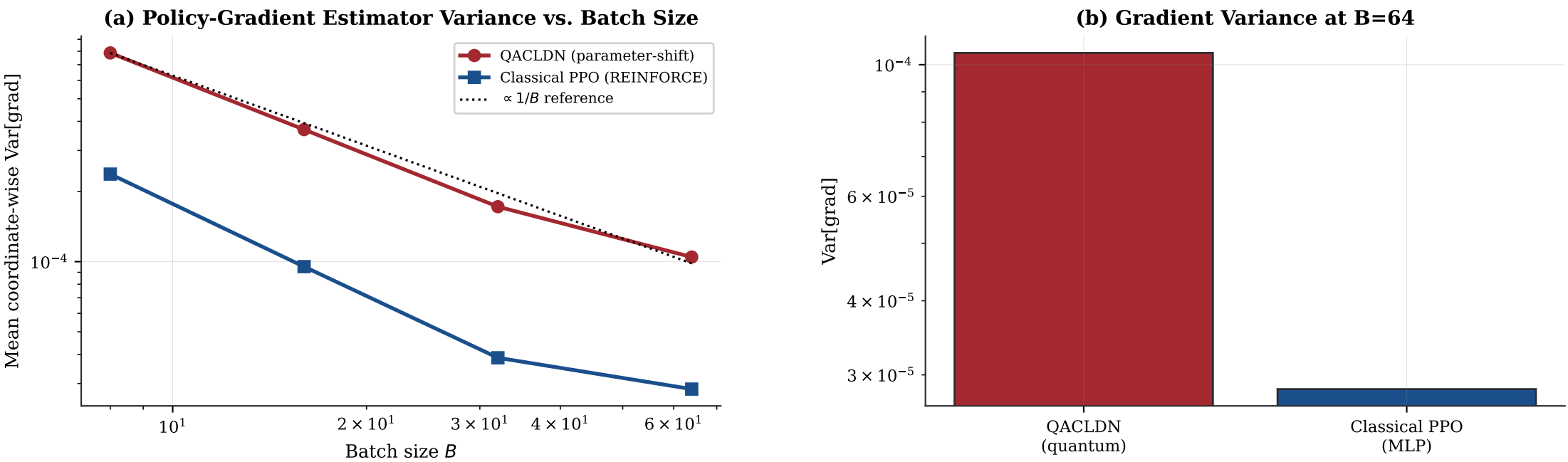


Figure 6 — Policy-Gradient Variance and Manuscript-Style Summary Table



(c) Trajectory-Quality Summary Table (real evaluation data, 60 episodes x 3 seeds)

Method	Params	SR [%]	CR [%]	PLR	EE	Mean Return
QACLDN	143	0.0 ± 0.0	68.3 ± 14.2	1.18	48.26	-1332.7
VQC-NE	143	0.0 ± 0.0	46.1 ± 6.1	1.12	48.62	-1343.6
Classical PPO	2057	0.0 ± 0.0	26.7 ± 2.7	0.67	52.16	-1493.6
Soft Actor-Critic	2057	0.0 ± 0.0	26.7 ± 2.7	0.66	52.47	-1502.2
Classical DQN	1072	0.0 ± 0.0	36.1 ± 10.0	0.93	53.72	-1470.1